

Forecasting Financial Risk Using Quantile Random Forests ^{*}

Robert James[†]

Jessica Wai Yin Leung[‡]

Abstract

This paper introduces a financial risk forecasting model that effectively exploits information from a large set of economic and financial predictor variables. The model is built using Generalized Quantile Random Forests, a non-parametric machine learning method that naturally permits variable interactions and non-linear relationships. We use a model-free variable screening technique and a robust cross-validation approach to minimize the risk of over-fitting. Our risk model produces competitive Value-at-Risk and Expected Shortfall forecasts at both a one-day ahead and a 10-day ahead horizon. A dynamic portfolio insurance strategy that uses the VaR and ES forecasts from our risk model generates attractive Sharpe, Sortino, and Omega ratios, particularly at the 10-day forecast horizon. Additionally, we provide a detailed analysis of the dynamic importance of our predictor variables. Our findings demonstrate the utility of combining large datasets with tree-based algorithms for financial risk forecasting.

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[†]Discipline of Business Analytics, University of Sydney, NSW 2006, Australia; email: r.james@sydney.edu.au

[‡]Department of Econometrics and Business Statistics, Monash University, VIC 3145, Australia; email: jessica.leung@monash.edu (corresponding author)

1 Introduction

Financial risk forecasting is a ubiquitous exercise. Accurate risk forecasts are essential for financial institutions to ensure compliance with capital adequacy requirements. Furthermore, these forecasts play a critical role in guiding quantitative portfolio construction and asset allocation. The Value-at-Risk (VaR) and the Expected Shortfall (ES) are principal risk measures and underpin international financial regulations (see, e.g. Basel Committee on Banking Supervision, 2016). Specifically, the VaR is the α -quantile of the distribution of returns on an asset or investment portfolio. The ES is the average of the worst returns beyond the α -quantile. In this study, we forecast the VaR and ES with a Generalized Quantile Random Forest (GQRF) model which effectively captures non-linear relationships and interactions among a large feature set.

Many existing financial risk forecasting models only use information embedded in historical asset prices to forecast the VaR and the ES. This is potentially a major shortcoming. For example, if prices are not fully efficient, and hence do not reflect all available information, then additional variables may be required to adequately forecast the dynamics of tail returns. Since the introduction of the efficient markets hypothesis, the literature has documented countless violations of the theory (Subrahmanyam, 2008; Lim and Brooks, 2011). More recently, an alternative theory, known as the adaptive markets hypothesis (AMH) (Lo, 2004, 2012), has been proposed to address the shortcomings of the traditional efficient markets hypothesis. In particular, the AMH considers that the degree of information efficiency is dynamic and depends upon prevailing market conditions, competition, and investor characteristics. The AMH implies that financial risk forecasting models should dynamically incorporate additional predictor variables as the degree of informational efficiency varies over the business cycle.

When financial risk forecasting models have incorporated additional variables, prior literature has typically only considered measures of equity market volatility (see, e.g. Gerlach and Wang, 2020; Opschoor and Lucas, 2021). Prior literature has largely ignored the possible benefits that could be realized by exploiting the wealth of economic and financial data that is now available at a high frequency. Moreover, existing models that allow for exogenous variables often restrict how these variables are related to financial risk (e.g. only allowing for a linear relationship), are unable to perform variable selection which can be important when the informational content of asset prices changes through time, do not easily allow for higher order variable interactions, and cannot successfully handle high dimensional data.

Our approach adapts the Generalized Random Forest introduced in Athey et al. (2019). The GQRF risk model that we develop naturally facilitates variable interactions, allows variables to explain non-linearities in financial risk, and handles a large number of predictor variables thereby alleviating problems associated with the curse of dimensionality in traditional quantile regression. Moreover, since our approach uses a non-parametric machine learning model, we make minimal assumptions about the data-generating process and instead learn the quantile that best fits current data. This is attractive because there is no consensus regarding the correct distribution of financial returns and this distribution may change over time. Hence, the GQRF model that we design addresses the limitations of existing financial risk forecasting models and allows practitioners to exploit a vast scale of economic and financial data to improve VaR and ES forecasts. In the case where some predictor variables are uninformative about future financial risk, the

dynamic variable selection mechanisms incorporated into our method can effectively ignore these irrelevant predictors. Importantly, this variable selection mechanism aligns well with the implications of the AMH regarding time-varying market efficiency and the role of additional predictor variables for financial risk forecasting.

When using any machine learning model, the training process must be designed to minimize over-fitting. This is particularly important in our risk forecasting application because financial returns are noisy and information about the tail quantile is naturally sparse. To mitigate the risk of over-fitting, we propose to first filter variables using a model-free variable screening technique (see, e.g. Borup et al., 2022; He et al., 2013). That is, at each forecast time we select only the subset of variables that are most strongly related to the target quantile to use in the GQRF model. Moreover, we use purged and embargoed randomized 5-fold cross-validation to select the hyper-parameters of our GQRF model. The purged and embargoed cross-validation process ensures that there is no information leakage between the training and testing sets (De Prado, 2018). These two approaches ensure that our GQRF model is sufficiently generalizable in low signal-to-noise financial risk forecasting environments (Timmermann, 2018).

As inputs to our GQRF model, we use over 100 economic and financial variables that can be broadly categorized into 5 groups; (i) variables derived from past prices/returns, (ii) measures of volatility, (iii) measures of liquidity, (iv) measures of economic and financial stress, and (v) variables engineered from other non-traditional data sources, such as Google Trends search volume data. The majority of these variables have not been considered for financial risk forecasting by prior literature. We benchmark our approach against popular parametric and semi-parametric financial risk forecasting models and find that our GQRF model is exceptionally competitive, often outperforming these benchmark models. In particular, we find that the VaR and ES forecasts from our GQRF model are only rejected from the 90% and 75% model confidence sets, that is the set of best forecasting models, twice for all stocks, horizons, and risk levels that we study. In contrast, all benchmark models are rejected numerous times from the model confidence sets. When used in a dynamic portfolio insurance trading strategy we find that the VaR and ES forecasts from our GQRF risk model generate attractive Sharpe, Sortino, MAR, and Omega ratios when compared to benchmark models, particularly at the 10-day ahead forecast horizon.

Our tree-based forecasting approach also allows us to examine the relative importance of the economic and financial predictor variables that we study. This analysis is important because it can inform practitioners and regulators about new sets of variables that could be used to monitor the dynamics of financial risk ex-ante. We find that multiple measures of volatility, both idiosyncratic and market-wide, are consistently important at the one-day forecast horizon. In particular, volatility and trading activity in the after-market-close trading session, measures of global equity market momentum and breadth, and a diverse set of financial stress indices are also important for both one-day ahead and 10-day ahead risk forecasting in our non-parametric model. More interestingly, we find that variables describing retail investor trading activity and trading activity in inverse ETFs are salient predictors of one-day ahead risk, particularly in recent years. Finally, variables that we engineer from Google Trends data and composite OECD leading indicators are particularly important for making accurate longer-term risk forecasts.

Traditional and widely used parametric approaches to forecast multi period VaR and ES include ARCH and GARCH type models of different frequencies (see e.g. Angelidis and Degiannakis, 2008; Degiannakis et al., 2014; Giot and Laurent, 2004; Degiannakis and Potamia, 2017). Other popular tools include the semi-parametric CaViaR type models of Engle and Manganelli (2004a); Taylor (2019), models that use extreme value theory such as the GARCH-EVT model of McNeil and Frey (2000) and the HAR-EVT model of Bee et al. (2016). We contribute to the literature by designing a machine learning approach to forecast the VaR and ES that can successfully exploit the benefits of data-rich environments by allowing for dynamic variable selection and non-linear variable interactions. This is important because recent work that applies machine learning models to financial data emphasizes the importance of these model attributes (see, e.g., Rasekhschaffe and Jones, 2019; Gu et al., 2020). To the best of our knowledge, we are the first to demonstrate the utility of tree-based methods for financial risk forecasting. Moreover, we believe our study analyzes the largest set of economic and financial variables for the task of financial risk forecasting in the literature. Our results highlight the dynamic importance of idiosyncratic and market-wide information and demonstrate the utility of combining many predictor variables for financial risk forecasting.

Naturally, our study is also related to an emerging literature that applies machine learning models, in particular random forests, for time series forecasting in economics and finance (see, e.g., Borup and Schütte, 2022; Gu et al., 2020; Medeiros et al., 2021). Recently Kelly et al. (2022) show both theoretically and empirically that high-dimensional machine learning models can provide substantial improvements over traditional approaches when forecasting financial returns. We outline the technicalities associated with applying a machine learning approach to forecast financial risk and confirm that such models are useful for this application. Future stock price movements are determined by the aggregate interactions of market participants, each possessing different information sets, statistical models, biases and heuristics, and exogenous information sets. As such, financial risk forecasting models should capture any non-linear relationships that arise in such environments. Moreover, the complexity of modern financial markets means that risk managers need to be using state-of-the-art techniques and contemporary datasets to anticipate extreme future price movements. Our study addresses these practical requirements.

The remainder of this paper is as follows. Section 2 introduces the GQRF model, outlines our approach to forecast the ES, describes the quantile adaptive variable screening procedure, and how we tune the model hyper-parameters. Section 3 describes our predictor variables and the sample used to backtest our GQRF risk model. The VaR and ES backtest results are presented in Section 4. Investment performance statistics from portfolio insurance strategies constructed using the VaR and ES forecasts are compared in Section 5. A discussion regarding the variable importance scores is presented in Section 6. Finally, Section 7 concludes.

2 Risk Forecasting Using Generalized Quantile Random Forests

Let $\mathcal{D} = \{(X_1, Y_1), \dots, (X_T, Y_T)\}$, $(X_t, Y_t) \in \mathcal{X} \times \mathbb{R}$, $\mathcal{X} \subseteq \mathbb{R}^p$ denote the data available for financial risk forecasting. In this context, daily negative returns of a financial asset are $r_t = -(\log(P_t) - \log(P_{t-1}))$ and $Y_t = \sum_{i=1}^h r_{t+i}$ where P_t is the price of a financial asset or investment portfolio at time t and h is the forecast

horizon. The cumulative distribution function of Y_t is $F_t(y) = P(Y_t \leq y)$. We are interested in making direct forecasts of the α -quantile of F_t for $\alpha \in (0, 1)$ using the information in X_t . This quantile is known as the VaR in financial contexts. Typically α is close to one, so VaR forecasting involves modeling the upper tail of F_t , that is extreme financial losses. When the discussion is unambiguous, we will suppress the time t , horizon h , and confidence level α scripts.

The GQRF model, introduced by Athey et al. (2019) constructs a non-parametric estimate of the conditional quantile by finding the solution to the following local estimating equation,

$$\widehat{VaR} = \hat{q}(x) = \arg \min_q \left\{ \left\| \sum_{t=1}^T a_t(x) \psi(y_t, q, \alpha) \right\|_2 \right\} \quad (1)$$

where $a_t(x)$ are weights that measure the relevance of the t^{th} training observation to the quantile at x and

$$\psi(y_t, q, \alpha) = (\alpha - \mathbb{1}(y_t < q)) y_t \quad (2)$$

is the quantile moment function (see, e.g., Koenker and Hallock, 2001). The adaptive weights $a_t(x)$ are constructed using an ensemble of tree-based neighborhoods derived from a random forest model, rather than from a deterministic kernel function or the K-Nearest Neighbor approach. As such, these weights benefit from all the advantages of random forests, namely their ability to handle high dimensional data, model non-linear functions and variable interactions, and effectively ignore irrelevant variables. Below we describe the approach of Athey et al. (2019) to grow decision trees that are specifically tailored to construct the adaptive weights for quantile estimation, then provide the equation used to compute the adaptive weights and some details regarding the construction of so-called “honest” trees.

Random forests, introduced by Breiman (2001), aggregate the predictions from $b = 1, \dots, B$ regression trees built using random sub-samples¹ of the data. Each tree is a partitioning estimate that separates the feature space into non-overlapping hyper-rectangles (nodes) via a recursive sequence of binary splits. Given a sub-sample of the data denoted by $\tilde{\mathcal{D}}$, a tree splitting algorithm seeks to recursively divide a parent node P into exactly two child nodes $C_1, C_2 \subseteq \tilde{\mathcal{D}}$ by making an axis aligned cut of the feature space. The location of this cut is chosen using a random sub-sample of predictor variables.

In our setting, we do not know the true conditional quantile for any choice of in-sample training data. This means that we cannot easily seek splits that minimize a measure of the prediction error, as in the standard CART splitting rule. Moreover, obtaining estimates of the conditional quantiles in each child node for every possible split by minimizing (2) is a computationally intensive task. Instead, Athey et al. (2019) propose to work with the following target criterion (see Proposition 1 of Athey et al. (2019)):

$$\Delta(C_1, C_2) := n_{C_1} n_{C_2} / n_P^2 (\hat{q}_{C_1}(\tilde{\mathcal{D}}) - \hat{q}_{C_2}(\tilde{\mathcal{D}}))^2 \quad (3)$$

¹The trees in a random forest can be constructed using either bootstrap samples or sub-samples of the data. However, we only consider sub-samples since this is the approach used in Athey et al. (2019) and is required in their theoretical analysis to achieve consistency.

where n_{C_1} and n_{C_2} are the number of observations in the child nodes C_1 and C_2 , and $\hat{q}_{C_1}$ and $\hat{q}_{C_2}$ are the estimated conditional quantiles in the child nodes. Choosing splits that maximize (3) would grow trees that increase the in-sample heterogeneity of quantile estimates. Such trees are specifically tailored for constructing the adaptive weights a_i , because these weights play a similar role to a kernel function in (1).

However, solving for $\hat{q}_{C_1}$ and $\hat{q}_{C_2}$ over all possible choices of split points is computationally expensive. Instead, Generalized Random Forests use gradient-based approximations of the conditional quantile in each of the child nodes. This gradient-based splitting algorithm proceeds by recursively applying the following two steps. First, we compute the conditional quantile in the parent node as the solution to the estimating equation, that is

$$\hat{q}_P = \arg \min_q \left\{ \left\| \sum_{\{t: X_t \in P\}} \psi(y_t, q, \alpha) \right\|_2 \right\} \quad (4)$$

Then, using the negated gradient of the expectation of the quantile moment function denoted by $A_P = \nabla E[\psi_{\hat{q}_P}(y_t) | X_t \in P]$, we compute the so-called pseudo responses as

$$\rho_t = -A_P^{-1} \psi(y_t, \hat{q}_P, \alpha) = \mathbb{1}(y_t > \hat{q}_P) \quad (5)$$

Second, we apply the CART splitting rule to split the parent node into two child nodes to maximize the following criterion

$$\tilde{\Delta}(C_1, C_2) = \sum_{\{i: X_i \in C_1\}} \rho_i^2 + \sum_{\{j: X_j \in C_2\}} \rho_j^2 \quad (6)$$

In summary, the GQRF model builds trees by first computing the α -quantile of the data in the parent node and then searching for an axis-aligned cut in the feature space that maximizes the number of observations in both child nodes above the α -quantile.

When estimating k different quantiles simultaneously, as will be required for our estimation of the ES discussed below, Athey et al. (2019) propose to first compute each quantile in the parent node by solving (4) to obtain the increasing sequence of quantiles $\hat{q}_{1,P} \leq \dots \leq \hat{q}_{k,P}$, and then label each observation according to which quantile interval it belongs to. The split point is chosen by minimizing the Gini loss.

After growing the forest, we can average the size of the local neighborhoods produced by the different trees to estimate the adaptive weight for any point x . Let $L_b(x)$ denote the set of observations that fall in the same terminal (“leaf”) node as x in tree b of the forest. Athey et al. (2019) define the adaptive weights as

$$a_t(x) = \frac{1}{B} \sum_{b=1}^B \frac{\mathbb{1}(X_t \in L_b(x))}{|L_b(x)|} \quad (7)$$

The idea is that each tree describes a small region of the feature space that is relevant for x and that the average of these different neighborhoods generates weights that give stable estimates of $\hat{q}(x)$.

We always grow so-called honest trees (see, e.g., Wager and Athey, 2018). Specifically, an honest tree is constructed by further splitting the random sub-sample of data $\tilde{\mathcal{D}}$, into two non-overlapping sub-samples, denoted by $\mathcal{I}_b$ and $\mathcal{J}_b$. The sub-sample $\mathcal{J}_b$ is only used for deciding splits and building the tree structure. The

sub-sample $\mathcal{I}_b$ is only used to define $L_b(x)$ and construct the adaptive weight $a_t(x)$. Honest trees naturally control for over-fitting because the observations used to construct the adaptive weights were “held-out” when constructing the tree. This is particularly useful when using random forests in low signal-to-noise environments, such as our financial risk forecasting application.

2.1 Forecasting the Expected Shortfall

The GQRF model is used to obtain an estimate of the VaR. In light of the recent Basel Accords (Basel Committee on Banking Supervision, 2016, 2017), we are also interested in forecasting the ES. When F_t is continuous and we are interested in the upper tail of extreme financial losses, as is always the case in our study, the ES is

$$ES_t^{(\alpha)}(Y_t|X_t) = \mathbb{E}[Y_t|Y_t \geq VaR_t^{(\alpha)}(Y_t|X_t)]. \quad (8)$$

To forecast the ES, we rely upon the following relationship (Acerbi and Tasche, 2002; Emmer et al., 2015)

$$ES_t^{(\alpha)} = \frac{1}{1-\alpha} \int_{\alpha}^1 VaR_t^{(\alpha)} d\alpha \quad (9)$$

$$\approx \frac{1}{M} \sum_{m=1}^M q_t^{(\alpha_m)}, \quad \alpha_m \geq \alpha \quad (10)$$

as $M \rightarrow \infty$. In other words, the ES can be approximated by averaging the tail VaRs beyond the α -quantile of F_t . Based upon this equivalence, we propose to combine multiple VaR forecasts from the GQRF model to construct a non-parametric forecast of the ES. This idea is closely related to the recent work of Storti and Wang (2022). Both Kratz et al. (2018) and Storti and Wang (2022) demonstrate that the choice of M is not crucial and that small values of M tend to work well for approximating the ES. With computational burdens in mind, and because data in the tails are scarce, we set $M = 4$ and construct a forecast of the ES by combining four VaR forecasts at equally spaced and increasing levels of confidence. Specifically, these levels of confidence are,

$$\alpha_m = \alpha + \frac{m-1}{M}(1-\alpha), \quad m = 1, \dots, M \quad (11)$$

While expression (10) implies equal weight for each of the four VaR forecasts, we choose to estimate optimal combination weights in a data driven manner. This is because our VaR estimates are themselves derived from an imperfect model so we expect that an optimal combination approach should allow the non-parametric ES forecast to better fit the empirical tail properties of the return distribution, and potentially adapt to any estimation error in individual VaR forecasts. To estimate weights we rely upon the result that the 2-dimensional vector $(VaR^{(\alpha)}, ES^{(\alpha)})$ is elicitable². Several strictly consistent joint scoring functions for the VaR and ES have been proposed (see, e.g., Nolde et al., 2017; Patton et al., 2019). In this study we minimize the following strictly consistent scoring function derived by Nolde et al. (2017) to estimate the

²Elicitability of the 2-vector $(VaR^{(\alpha)}, ES^{(\alpha)})$ means that there exists at least one strictly consistent scoring function which is minimized by the true pair of forecasts $(VaR^{(\alpha)}, ES^{(\alpha)})$ in expectation.

combination weights³,

$$S(VaR^{(\alpha)}, ES^{(\alpha)}, y, \alpha) = \mathbb{1}(y_{t,h} > VaR^{(\alpha)}) \frac{y - VaR^{(\alpha)}}{2\sqrt{ES^{(\alpha)}}} + (1 - \alpha) \frac{VaR^{(\alpha)} + ES^{(\alpha)}}{2\sqrt{ES^{(\alpha)}}} \quad (12)$$

Specifically, we apply the following affine approximation of the ES introduced by (Storti and Wang, 2022):

$$\widehat{ES}_t^{(\alpha)} = w_0 + \sum_{m=1}^M w_m \widehat{VaR}_t^{(\alpha_m)} \quad (13)$$

where

$$\begin{aligned} \widehat{\mathbf{w}} = \arg \min_{\mathbf{w}} \quad & \sum_{t=1}^T S(\widehat{VaR}_t^{(\alpha)}, \widehat{ES}_t^{(\alpha)}, y_t, \alpha) \\ \text{s.t.} \quad & w_m \geq 0 \quad \forall \quad m = 0, \dots, M \end{aligned} \quad (14)$$

and $\mathbf{w} = (w_0, \dots, w_{M+1})$. An intercept (w_0) is included to account for any bias introduced by the discretized approximation of the ES.

2.2 Variable Screening

Motivated by the well-known presence of a low signal-to-noise ratio in financial forecasting applications (see, e.g., Timmermann, 2018), we propose to first select only a small subset of the most relevant variables to use as inputs to the GQRF model. When the true model only uses a sparse set of predictor variables, as is often the case in finance and economics (see, e.g. Chinco et al., 2019; Yousuf and Feng, 2022)), the recent work of Borup et al. (2022), demonstrates that the random forest tuning parameter which controls the number of feasible splitting variables *mtry* is unlikely to be sufficient to ensure individual decision trees place splits among important variables. In particular, the Borup et al. (2022) advocate for selecting the most important 5%-30% of predictors when using random forest models to forecast economic and financial time series⁴. While a variable screening step is expected to improve the forecast accuracy of our GQRF risk model and reduce the risk of over-fitting, it is also of practical importance because it considerably reduces the computational cost of fitting the model by reducing the size of the feature space.

Let

$$\mathcal{A} = \{j : q(x) \text{ functionally depends upon } X_j, j = 1, \dots, p\}$$

denote the set of so-called active variables (see, e.g., Mai and Zou, 2015). Variable screening methods assume that $|\mathcal{A}| < p$ and seek to conservatively discover the set of variables $\mathcal{S}$ such that $\mathcal{A} \subset \mathcal{S}$. Borup et al. (2022) demonstrate that screening variables increases the probability of splitting on important predictors and always improves the quality of individual decision trees in the forest⁵.

³Our results are qualitatively robust to using the scoring function introduced by Patton et al. (2019).

⁴This approach is also closely related to the work of Bai and Ng (2008)

⁵However, the trade-off is that trees can become significantly more correlated so that the random forest may not reduce variance as much.

To identify $\mathcal{S}$, we use the model free quantile adaptive screening procedure of He et al. (2013), which is based on the idea that

$$Y \text{ and } X_j \text{ are independent} \quad \Leftrightarrow \quad \hat{v}_j = \hat{q}^{(\alpha)}(x) - \hat{q}^{(\alpha)} = 0 \quad \forall \alpha \in (0, 1)$$

When the difference between the conditional α -quantile and the unconditional α -quantile, is zero for all risk levels, X_j , the j^{th} predictor variable, contributes no additional information about the log return. That is, X_j is independent of Y .

The conditional α -quantile term $\hat{q}(x)$ is approximated non-parametrically using b-spline quantile regression. The unconditional α -quantile term $\hat{q}$ is computed using the empirical distribution function. He et al. (2013) show that a variable screening approach that selects important predictors using the $\hat{v}_j$ statistic enjoys the sure screening property. That is, the set of selected variables contains all truly important variables with probability 1 as the sample size T approaches infinity. Moreover, the approach allows covariates to be non-linearly related to the α -quantile which aligns nicely with the hypothesized benefits of the GQRF model for financial risk forecasting. In practice, variables are ranked based upon the value of $\|\hat{v}\|_2^2$ we retain the top 20% of variables⁶.

2.3 Hyper-parameter Tuning

Traditionally, the hyper-parameters in the random forest model are selected to minimize the so-called out-of-bag error prediction error. However, because we make extensive use of lag operators and moving averages in our feature engineering process, the randomized sub-sampling mechanism of the GQRF model will lead to information leakage across the training data and the out-of-bag data. Moreover, information leakage between the training data and the out-of-bag data will occur because the target quantile for multistep ahead forecasting is based upon cumulative h -period ahead forward returns.

To avoid these two types of information leakage, we use purged and embargoed randomized K -fold cross-validation⁷, introduced in De Prado (2018). This approach first splits the training data into K contiguous folds. Then we purge any observations in the training folds where there is known information overlap with the observations used to compute h -period ahead returns in the test fold. For example, if $K = 5$ and we are using the third fold as the test set, we would like to purge any observations at the end of the second training fold where the information sets used to compute h -period ahead returns overlap. Additionally, we also remove, a fixed number of observations from the beginning of the training fold that immediately follows the test fold to alleviate information leakage associated with serial correlation in predictor variables, a process referred to as embargoing by De Prado (2018). We set $K = 5$ and choose to embargo 22 daily

⁶Although we do not do so, the number of variables retained could be a hyper-parameter and tuned using the K -fold cross-validation approach we describe in Section 2.3

⁷ K -fold cross-validation assumes that the hyper-parameters are constant over the training dataset. While this may be a slightly restrictive assumption, we prefer K -fold cross-validation to time series cross-validation for example, because the latter is not computationally feasible in practice when performing rolling/expanding window time series forecasting with machine learning models. However, a benefit of the K -fold cross-validation approach is that since we only choose a single set of hyper-parameters and hence we are less likely to overfit the training data.

observations, since this is the size of the largest lag operator that we use.

We search for the optimal value of the minimum leaf size in the set $\{5, 10, 25, 50\}$, the optimal subsampling fraction in $\{0.3, 0.4, 0.5, 0.7\}$, the honesty fraction, that is the fraction of subsampled observations used to determine splits, in $\{0.3, 0.5, 0.7\}$ and the value of a split imbalance penalty parameter in $\{0.05, 0.1, 0.2\}$. The number of trees is fixed at 2000, and the number of features used to search for splits, often referred to as the *mtry* hyper-parameter, is set to $\sqrt{p}$. We sample 100 random combinations of the hyper-parameter search space. The size of the grids of candidate hyperparameters and the number of randomized trials are purposely kept at a modest size to limit the likelihood that we overfit the training data.

Algorithm 1 presents the pseudo-code summarizing the steps involved in our GQRF financial risk forecasting model. We perform the quantile adaptive variable screening step and compute the optimal combination weights used to forecast the expected shortfall each day when a new forecast is required. This ensures that the algorithm can quickly adapt to changing informational environments. The hyper-parameters of our GQRF model are only re-tuned at the beginning of every month.

Algorithm 1: GQRF Financial Risk Forecasting Model

- Input** : $(\mathbf{X}, \mathbf{y}), \alpha^{(VaR)}, \alpha^{(ES)}, \boldsymbol{\theta}^{(VaR)}, \boldsymbol{\theta}^{(ES)}$
Output : $\widehat{VaR}_t^{(\alpha)}, \widehat{ES}_t^{(\alpha)}, \mathbf{w}$
- 1 Construct $\mathcal{S}$ for the VaR and ES models using quantile adaptive screening (see Section 2.2). Denote $\tilde{\mathbf{X}}^{(VaR)}$ and $\tilde{\mathbf{X}}^{(ES)}$ the training data containing the screened variables.
 - 2 **if** *is_month_start* **then**
 - 3 Tune hyper-parameters using purged and embargoed K -fold CV and obtain $\boldsymbol{\theta}^{(VaR)}$ and $\boldsymbol{\theta}^{(ES)}$ (see Section 2.3).
 - 4 Fit the GQRF model using $(\tilde{\mathbf{X}}^{(VaR)}, \mathbf{y})$ and $\boldsymbol{\theta}^{(VaR)}$ to forecast $VaR^{(\alpha)}$.
 - 5 Fit the GQRF model using $(\tilde{\mathbf{X}}^{(ES)}, \mathbf{y})$ and $\boldsymbol{\theta}^{(ES)}$ to estimate $VaR^{(\alpha_1)}, \dots, VaR^{(\alpha_M)}$ simultaneously.
 - 6 **if** $\mathbf{w} = \{\}$ **then**
 - 7 Find starting values for the ES combination weights (see Section 3).
 - 8 Estimate the optimal combination weights $\hat{\mathbf{w}}$ by solving (14), using $\mathbf{w}$ as starting values. Update $\mathbf{w} = \hat{\mathbf{w}}$.
 - 9 Forecast $ES^{(\alpha)}$ using $VaR^{(\alpha_1)}, \dots, VaR^{(\alpha_M)}$ and $\hat{\mathbf{w}}$ following eq (13)

As inputs, our GQRF risk model takes a vector of h -period ahead log returns $\mathbf{y}$, a matrix of predictor variables $\mathbf{X}$, the level of confidence α , and the hyper-parameters of the VaR and ES GQRF risk models. The algorithm returns the VaR and ES forecasts as well as the vector of weights obtained from minimizing expression (14).

3 Data

We backtest the VaR and ES forecasts from our GQRF model using a random sample of 10 U.S. equities in the S&P100 index. The ticker symbols for each equity we study are; AAPL, MSFT, JPM, IBM, GE, UNH, INTC, CSCO, ORCL, and MRK. For each equity, we obtain stock price data from Compustat for the period 2003-09-10 to 2019-11-29 and make 2250 out-of-sample risk forecasts for the period 2010-12-22 to 2019-11-29. This sample period is governed by the availability of the predictor variable data that we

describe below. We forecast the VaR for $\alpha \in \{0.95, 0.975, 0.99\}$, the ES for $\alpha \in \{0.95, 0.975\}$. Risk forecasts are computed for both the one-day ahead ($h = 1$) and 10-day ahead ($h = 10$) horizons since multi-step ahead forecasts are particularly important for practitioners. Specifically, to produce the 10-day ahead forecasts we fit our GQRF risk model to the rolling 10-day ahead log return time series. For each level of α , we train a distinct GQRF risk model thereby, allowing the dynamics of risk to vary for different locations of the tail. To avoid look-ahead bias when training our GQRF model over the backtests we drop the most h recent training observations at each forecast time t since y is the cumulative h -period ahead return in our context.

As benchmarks, we estimate a number of popular parametric and semi-parametric risk models from prior literature. First, we estimate a parametric GARCH(1,1) model with Skew- t error (Hansen, 1994). Second, we use the GARCH-EVT model of McNeil and Frey (2000), a semi-parametric model that applies the peaks-over-threshold extreme value theory to approximate the tail distribution of the standardized residuals from the GARCH model. We use a GJR-GARCH(1,1) specification for this benchmark risk model since Trapin (2018) demonstrate that GARCH models which incorporate the so-called leverage effect are necessary to remove dependence among the most extreme returns. Third, we implement a HAR Range-EVT model which combines a volatility model with the peaks-over-threshold extreme value theory to forecast financial risk. We use the daily range volatility estimator of Garman and Klass (1980) within the HAR Range-EVT model, as its calculation only requires daily open, high, low and close price data that is readily available to practitioners. Moreover, Lyócsa et al. (2021) find that a HAR models which use daily range volatility estimators are competitive volatility forecasting models.

Our final set of benchmark models are the semi-parametric CaViaR and CaViaR-EVT models (see, e.g., Engle and Manganelli, 2004a,b), and AL-CaViaR (Asymmetric Laplace CaViaR) of Taylor (2019). Each of these CaViaR type models uses the so-called asymmetric slope type specification. To obtain an estimate of the ES from the CaViaR model we compute the sum of the estimated VaR and the average of all returns that exceed this VaR, a method known as the mean exceedence approach (see, e.g., Taylor, 2019). The implementation of each of these models is described in Appendix A.1 - A.3.

A rolling window of 1500 trading days is used to train each risk forecasting model. We prefer the rolling window forecasting scheme because this limited memory estimator can provide a better local approximation when the target series is auto-correlated and/or the prediction model is potentially misspecified (see, e.g. Giacomini and White, 2006). To set the starting values used in the numerical optimization routine in the first rolling training window for estimating the optimal combination weights in our ES forecasting approach, we draw 10^3 parameter vectors from a Dirichlet distribution and record the 10 weight vectors that give the best values of the objective function. We then use these 10 weight vectors as initial values in the numerical optimization routine and select the single best weight vector that minimizes the objective function. For all subsequent moving windows, we use the estimated weight vector from the previous window as starting values.

Table 8 in Appendix B lists the predictor variables that we use in our GQRF risk model together with their associated data source and frequency information. A more detailed description of each variable is provided in the online supplementary materials file. The variables that we include in our GQRF model can

be broadly categorized into variables that describe past return/prices, financial market volatility, liquidity, and economic and financial conditions. We also include a handful of variables derived from non-standard data sources, such as variables engineered from Google Trends internet search volume data and OECD leading indicator data. To the best of our knowledge, this is among the largest set of predictor variables considered for financial risk forecasting.

Since some variables are only available at a weekly or monthly frequency, each variable is constructed to ensure that there is no look-ahead bias at forecast time t . For example, when we know the specific release date for weekly or monthly variables, such as the Chicago Fed National Financial Conditions Index, we forward fill the data between each specific release date. When we do not know the specific release date for weekly or monthly frequency variables we always use the 1-period lagged variable at any given week or month. When using lags of variables we typically restrict ourselves to using the 1-day, 5-day, and 22-day lags, which approximately correspond to one trading day, one trading week, and one trading month. However, our method allows practitioners to specify any specific lag/s of variables in the model.

Table 1 reports the first four sample moments of the daily stock return time series for each stock that we study together with the test statistic for the finite moments test introduced by Trapani (2016), and further refined by Degiannakis et al. (2023). We use a AR(1)-GARCH(1,1) to remove auto-correlation and conditional heteroskedasticity from the daily stock return time series and estimate the test statistic using the standardized residuals from this model. We compute the test statistic at the 95% level of confidence and use the same specifications as in Degiannakis et al. (2023). We find that the first three moments are finite for all stocks that we study. However, there is evidence to suggest that MSFT, CSCO, ORCL, and MRK have an infinite fourth moment. This evidence is broadly consistent with Trapani (2016). We expect that the bagging process inherent to random forest models combined with our robust hyper-parameter tuning process will be particularly beneficial in the presence of infinite higher order moments.

Table 1: Sample Moments

The Table reports summary statistics and the test statistic $Q^{(k)}(0.05)$ of Degiannakis et al. (2023) for the k^{th} moment at the 95% level of confidence. The test is conducted using the standardized residuals from an AR(1)-GARCH(1,1) model fit to the daily stock return time series. Asterisks indicate that the null hypothesis that the k^{th} sample moment is finite is not rejected.

	Mean	$Q^{(1)}(0.05)$	Std.Dev	$Q^{(2)}(0.05)$	Skewness	$Q^{(3)}(0.05)$	Kurtosis	$Q^{(4)}(0.05)$
AAPL	0.132	0.181	2.013	0.266	0.072	0.353	2.231	0.784
MSFT	0.043	0.189	1.535	0.324	-0.156	0.509	4.086	0.93*
JPM	0.034	0.179	2.118	0.256	0.293	0.321	8.969	0.702
IBM	0.008	0.179	1.285	0.292	-0.428	0.462	3.417	0.901
GE	-0.024	0.179	1.794	0.266	-0.144	0.380	6.154	0.864
UNH	0.059	0.179	1.763	0.266	-0.150	0.364	4.613	0.819
INTC	0.025	0.178	1.755	0.259	-0.195	0.388	2.786	0.868
CSCO	0.023	0.189	1.710	0.345	-0.393	0.583	4.365	0.943*
ORCL	0.034	0.186	1.670	0.326	-0.107	0.565	3.930	0.945*
MRK	0.015	0.180	1.474	0.312	-0.433	0.531	5.299	0.933*

4 Forecast Results

To evaluate the quality of the VaR forecasts we use the unconditional coverage (LR_{uc}), independence (LR_{ind}), and conditional coverage (LR_{cc}) tests of Christoffersen (1998), the Dynamic Quantile (DQ) test of Engle and Manganelli (2004a) with four lags, and the one sided test for VaR underestimation ($ZBWW_{uc}$) proposed by Ziggel et al. (2014). For each of these tests we report the number of times the test null hypothesis was rejected at the 5% level of significance over all ten stocks in our sample. We also report *Violation %* which is the fraction of VaR violations averaged over each of the 10 stocks. This statistic measures the distance from the target VaR violation percentage and positive values risk under-estimation while negative deviations indicate risk over-estimation. The aforementioned tests assess whether VaR forecasts are properly calibrated for the given level of confidence.

The quality of ES forecasts is evaluated using the mean exceedance residual test of McNeil and Frey (2000) (*McNeil Bootstrap*). We also compute the two-sided (*ESR Two-Sided*), one-sided (*ESR One-Sided*) and auxiliary (*ESR Auxiliary*) regression based tests of Bayer and Dimitriadis (2022). The two-sided ESR test is also referred to as the “strict ESR” test by Bayer and Dimitriadis (2022). Both the exceedance residual test and the one-sided regression test, also known as the “intercept ESR” test by Bayer and Dimitriadis (2022), assess whether a risk model systematically underestimates the ES. For each of these ES forecast tests we report the number of times the test was rejected at the 5% level of significance over all 10 stocks in our backtest. For both the VaR and ES forecasts We also report the total number of rejections across all statistical tests and stocks (*Total*) as a single summary measure of forecast performance.

The value of the quantile loss function ($S(VaR, y, \alpha)$) (see, e.g., Gneiting, 2011) and the value of the joint VaR and ES scoring function ($S(VaR, ES, y, \alpha)$) of Nolde et al. (2017) averaged over all 10 stocks is also reported. These scoring functions are particularly well suited for comparative risk model back-testing. Finally, for both the VaR and ES forecasts we report the number of times each model was rejected from the 75% (MCS 75%) and 90% (MCS 90%) model confidence sets (see, e.g. Hansen et al., 2011). The model confidence set identifies a group of models that will include the best forecasting model with a pre-specified degree of confidence while controlling the familywise error rate ⁸. The stationary block bootstrap size in this test is set using the algorithm described by Patton et al. (2009) and Politis and White (2004).

Table 2 reports the results for the one-day ahead VaR forecasts. The VaR forecasts from our GQRF risk model have an average violation % statistic that is very close to the target level of confidence and marginally overestimate the one-day ahead VaR across all three levels of confidence. Arguably, risk over-estimation is more preferable to risk under-estimation from a regulatory perspective. The benchmark semi-parametric CaViaR, CaViaR-EVT and AL-CaViaR models tends to underestimate the VaR at $\alpha = 0.975$ and $\alpha = 0.99$. The total number of rejections associated with our GQRF model is typically close to the smallest number of rejections achieved by the best benchmark forecasting models. This result demonstrates that the VaR forecasts from our GQRF risk model are generally well calibrated and that our approach is competitive to the single best benchmark model across all three levels of confidence.

The mean value of $S(VaR_t, y_t)$, the quantile loss function, associated with our GQRF risk model is

⁸Our results are robust to using the loss function studied by Patton et al. (2019) to construct the model confidence sets

Table 2: One-day Ahead VaR Forecast Results

The table reports the backtest results for one-day ahead VaR forecasts. The column *Violation %* reports the average fraction of VaR violations minus and measures risk over/under estimation. LR_{uc} , LR_{ind} , and LR_{cc} are the sum of rejections at the 5% level of significance for the unconditional coverage, independence and conditional coverage tests of Christoffersen (1998) respectively. $ZBWW_{uc}$ is the number of rejections at the 5% level of significance from the one sided test for VaR underestimation (Ziggel et al., 2014). DQ is the number of rejections at the 5% level of significance from the Dynamic Quantile test of (Engle and Manganelli, 2004a). *Total* is the sum of all rejections for the VaR calibration tests. $S(VaR, y)$ reports the mean value of the quantile scoring function averaged over all 10 stocks. *MCS 75%* and *MCS 90%* are the number of times each model was rejected from the 75% and 90% VaR model confidence sets.

Panel A: $\alpha = 0.95$										
Risk Model	Violation %	LR_{UC}	$ZBWW_{UC}$	LR_{Ind}	LR_{CC}	DQ	Total	$S(VaR, y, \alpha)$	MCS 75%	MCS 90%
GQRF	4.6132	2	2	1	1	0	6	3.1523	0	0
GJR-GARCH Skew-t	4.4623	4	3	1	5	2	15	3.3533	6	6
GARCH-EVT	4.1688	4	4	3	4	5	20	3.4015	9	9
HAR Range-EVT	4.9733	1	1	7	6	6	21	3.3343	5	5
CaViaR	4.8487	0	0	1	1	2	4	3.3129	3	3
CaViaR-EVT	4.8977	1	1	1	2	3	8	3.3215	3	3
AL-CaViaR	4.8089	0	0	1	0	1	2	3.3191	3	3
Panel B: $\alpha = 0.975$										
Risk Model	Violation %	LR_{UC}	$ZBWW_{UC}$	LR_{Ind}	LR_{CC}	DQ	Total	$S(VaR, y, \alpha)$	MCS 75%	MCS 90%
GQRF	2.2578	1	0	1	1	1	4	3.8399	0	0
GJR-GARCH Skew-t	2.3912	1	0	1	0	1	3	4.1799	4	4
GARCH-EVT	2.2043	1	1	1	2	3	8	4.2401	8	8
HAR Range-EVT	2.6355	0	1	6	5	4	16	4.1568	3	3
CaViaR	2.5867	0	0	0	0	1	1	4.1619	3	3
CaViaR-EVT	2.5779	1	1	2	1	2	7	4.1525	3	3
AL-CaViaR	2.5334	0	0	0	0	0	0	4.1597	3	3
Panel C: $\alpha = 0.99$										
Risk Model	Violation %	LR_{UC}	$ZBWW_{UC}$	LR_{Ind}	LR_{CC}	DQ	Total	$S(VaR, y, \alpha)$	MCS 75%	MCS 90%
GQRF	0.8446	1	1	0	1	1	4	4.9243	0	0
GJR-GARCH Skew-t	1.0401	1	1	0	0	0	2	5.5228	3	3
GARCH-EVT	0.9378	0	0	0	1	2	3	5.5746	4	5
HAR Range-EVT	1.1689	1	2	2	3	4	12	5.493	2	2
CaViaR	1.1822	1	1	1	1	3	7	5.5505	2	2
CaViaR-EVT	1.0934	1	1	0	1	1	4	5.5279	2	2
AL-CaViaR	1.1378	0	1	0	0	3	4	5.5775	2	2

substantially smaller than the value achieved by all benchmark models over all three levels of confidence that we study. This result suggests that the one-day ahead VaR forecasts produced by our GQRF model are often significantly closer to the true unobserved conditional quantile. Subsequently, VaR forecasts from our GQRF model are never rejected from either the 90% or the 75% VaR model confidence sets at any level of α for any stock in our backtest. In contrast, all benchmark models are rejected multiple times from both model confidence sets. Taking into account both rejections from the statistical calibration tests and rejections from the two model confidence sets, the one-day ahead VaR forecasts from our GQRF risk model are always associated with the smallest number of aggregate rejections.

Table 3 reports the results for the one-day ahead ES forecasts. When considering the only the first three ES backtests, which are strict tests of the ES, our GQRF risk model is particularly competitive at the $\alpha = 0.975$ level which is the level requested by regulators. Based upon the relatively small number of rejections associated with the *McNeil Bootstrap* and *ESR one-Sided* tests, we conclude that our GQRF risk model does not systematically underestimate the ES. Interestingly, the ES forecasts from the AL-CaViaR model perform very well compared to all risk models that we study. The mean value of $S(q^{(\alpha)}, ES^{(\alpha)}, y, \alpha)$, the joint VaR and ES scoring function, associated with our GQRF model is considerably smaller than scores associated with any benchmark model, suggesting that when considered together pairs of VaR and ES of point forecasts issued by our model tend to be more accurate than competing benchmark approaches. Finally, our GQRF risk model is never rejected from either model confidence set.

Table 3: One-day Ahead ES Forecast Results

The table reports the backtest results for one-day ahead ES forecasts. The column *McNeil Bootstrap* reports the number of times the mean exceedance residual test of McNeil and Frey (2000) was reject at the 5% level of significance. *ESR One-Sided*, *ESR Two-Sided* and *ESR Auxiliary* are the number of rejections at the 5% level of significance from the one-sided, two-sided and auxiliary regression based tests of Bayer and Dimitriadis (2022). *Total* is the sum of all rejections for the ES calibration tests. $S(VaR, ES, y, \alpha)$ reports the mean value of the joint VaR and ES scoring function (12). *MCS 75%* and *MCS 90%* are the number of times each model was rejected from the 75% and 90% ES model confidence sets.

Panel A: $\alpha = 0.95$								
Risk Model	McNeil Bootstrap	ESR Two-Sided	ESR One-Sided	ESR Auxiliary	Total	$S(VaR, ES, y, \alpha)$	MCS 75%	MCS 90%
GQRF	1	5	2	4	12	1.7469	0	0
GJR-GARCH Skew-t	8	2	2	3	15	1.8083	3	3
GARCH-EVT	2	2	3	2	9	1.823	8	8
HAR Range-EVT	2	1	1	1	5	1.8022	3	3
CaViaR	0	2	2	3	7	1.7983	3	3
CaViaR-EVT	10	3	6	2	21	1.8032	3	3
AL-CaViaR	1	0	0	0	1	1.7978	3	3
Panel B: $\alpha = 0.975$								
Risk Model	McNeil Bootstrap	ESR Two-Sided	ESR One-Sided	ESR Auxiliary	Total	$S(VaR, ES, y, \alpha)$	MCS 75%	MCS 90%
GQRF	0	2	1	7	10	1.9279	0	0
GJR-GARCH Skew-t	8	3	2	4	17	2.0273	4	4
GARCH-EVT	1	1	1	2	5	2.0399	6	7
HAR Range-EVT	4	0	0	1	5	2.0184	3	3
CaViaR	0	1	0	1	2	2.0216	3	3
CaViaR-EVT	7	1	2	1	11	2.0219	3	3
AL-CaViaR	1	0	0	0	1	2.02	3	3

Table 4 reports the results for the 10-day ahead VaR forecasts. We only focus on results for the LR_{uc}

and $ZBWW_{uc}$ unconditional coverage tests for this longer horizon. We find that our GQRF risk model almost always has an average violation % that is significantly closer to the target level of confidence at the 10-day horizon. Moreover, our GQRF model always has the smallest number of rejections for the unconditional coverage tests for any choice of α . This result demonstrates that our GQRF risk model does not systematically underestimate the VaR over longer horizons. At the $h = 10$ day horizon, the mean value of the quantile loss function associated with our QGRF model is often close to the value achieved by the best risk forecasting model but is no longer unanimously the smallest. Subsequently, we find that all risk models are generally included in the model confidence set, suggesting that it is more difficult to identify the best model using a consistent scoring function at longer forecasting horizons. Importantly, the value of $S(VaR_t, y_t)$ for our GQRF model is always smaller than the value achieved by all three benchmark semi-parametric CaViaR models, as this is the most similar model to the GQRF model.

Table 5 reports the results for the 10-day ahead ES forecasts. The ES forecasts from our GQRF risk model are associated with the second smallest number of calibration test rejections at the $\alpha = 0.975$ level. Moreover, the mean value of the scoring function $S(q^{(\alpha)}, ES^{(\alpha)}, y, \alpha)$ achieved by our GQRF model is competitive with benchmark models, particularly at the $\alpha = 0.975$ level. While the ES forecasts from the AL-CaViaR model performed well at the one-day ahead forecast horizon, they appear to be less competitive for longer horizon tail risk forecasting tasks.

Tables 2 - 5 show that the VaR and ES forecasting performance of benchmark risk models tends to fluctuate, depending on the choice of α and the forecast horizon h . In contrast, the VaR and ES forecast quality statistics associated with our GQRF risk model are almost always close to the value achieved by the best forecasting model. Taken together, backtest results suggest that the one-day ahead VaR and ES forecasts produced by our GQRF model are competitive, but also that our QGRF risk model generalizes well across the different choices of the confidence level α and the forecast horizon h for financial risk forecasting tasks.

5 Investment Strategy

To evaluate the economic gains that could be realized by using our GQRF risk model, we backtest a Dynamic Proportion Portfolio Insurance (DPPI) trading strategy (see, e.g. Happersberger et al., 2020). This strategy dynamically adjusts the exposure between a risky asset and a low-risk asset using a risk forecast to target a minimum level of capital at the end of a given investment horizon. At the beginning of each investment period, denoted by t^* (e.g. each year or each quarter), an investor sets a minimum portfolio value floor F_{t^*} that should be preserved at the end of the investment period. Then, the risk capital is calculated as the difference between the current value of the portfolio V_t and the net present value of the floor, also known as the discounted floor, that is

$$C_t = V_t - NPV_t(F_{t^*}) \quad (15)$$

To avoid a breach of the discounted floor the investment exposure to the risky asset is set such that the risk capital at time t is at least as high as the largest expected drop in the portfolio value between

Table 4: 10-day Ahead VaR Forecast Results

The table reports the backtest results for ten-day ahead VaR forecasts. The column *Violation %* reports the average fraction of VaR violations and measures risk over/under estimation. LR_{uc} is the sum of rejections at the 5% level of significance for the unconditional coverage test of (Christoffersen, 1998). $ZBWW_{uc}$ is the number of rejections at the 5% level of significance from the one sided test for VaR underestimation (Ziggel et al., 2014). *Total* is the sum of all rejections for the VaR calibration tests. $S(VaR, y)$ reports the mean value of the quantile scoring function averaged over all 10 stocks. *MCS 75%* and *MCS 90%* are the number of times each model was rejected from the 75% and 90% VaR model confidence sets.

Panel A: $\alpha = 0.95$							
Risk Model	Violation %	LR_{UC}	$ZBWW_{UC}$	Total	$S(VaR, y, \alpha)$	MCS 75%	MCS 90%
GQRF	4.9334	4	4	8	10.8405	1	1
GJR-GARCH Skew-t	4.4489	8	8	16	10.759	1	1
GARCH-EVT	2.8133	9	9	18	11.2358	5	5
HAR Range-EVT	4.4933	8	8	16	10.6116	0	0
CaViaR	6.7734	6	6	12	10.8624	0	0
CaViaR-EVT	6.8267	7	7	14	10.8623	1	1
AL-CaViaR	6.6	6	6	12	10.8128	1	1
Panel B: $\alpha = 0.975$							
Risk Model	Violation %	LR_{UC}	$ZBWW_{UC}$	Total	$S(VaR, y, \alpha)$	MCS 75%	MCS 90%
GQRF	2.7245	3	2	5	13.079	0	0
GJR-GARCH Skew-t	3.4088	5	5	10	12.9867	0	0
GARCH-EVT	1.7821	8	8	16	13.161	1	1
HAR Range-EVT	2.9955	6	6	12	12.7877	0	0
CaViaR	3.8932	7	7	14	13.2735	1	1
CaViaR-EVT	3.8754	7	7	14	13.303	0	0
AL-CaViaR	3.7244	6	6	12	13.1948	0	0
Panel C: $\alpha = 0.99$							
Risk Model	Violation %	LR_{UC}	$ZBWW_{UC}$	Total	$S(VaR, y, \alpha)$	MCS 75%	MCS 90%
GQRF	1.2533	2	2	4	16.2375	0	0
GJR-GARCH Skew-t	2.5333	9	9	18	17.4063	0	0
GARCH-EVT	1.0266	4	4	8	16.0281	0	0
HAR Range-EVT	1.8222	6	8	14	16.1717	0	0
CaViaR	1.6844	7	7	14	16.2859	0	0
CaViaR-EVT	1.7333	6	6	12	16.3882	1	1
AL-CaViaR	1.6934	9	9	18	16.3343	1	1

Table 5: 10-day Ahead ES Forecast Results

The table reports the backtest results for 10-day ahead ES forecasts. The column *McNeil Bootstrap* reports the number of times the mean exceedance residual test of McNeil and Frey (2000) was reject at the 5% level of significance. *ESR One-Sided*, *ESR Two-Sided* and *ESR Auxiliary* are the number of rejections at the 5% level of significance from the one-sided, two-sided and auxiliary regression based tests of Bayer and Dimitriadis (2022). *Total* is the sum of all rejections for the VaR calibration tests. $S(VaR, ES, y, \alpha)$ reports the mean value of the joint VaR and ES scoring function (12). *MCS 75%* and *MCS 90%* are the number of times each model was rejected from the 75% and 90% ES model confidence sets.

Panel A: $\alpha = 0.95$								
Risk Model	McNeil Bootstrap	ESR Two-Sided	ESR One-Sided	ESR Auxiliary	Total	$S(VaR, ES, y, \alpha)$	MCS 75%	MCS 90%
GQRF	5	10	3	9	27	3.2422	1	1
GJR-GARCH Skew-t	10	8	5	8	31	3.2185	1	1
GARCH-EVT	6	10	9	9	34	3.2833	3	3
HAR Range-EVT	9	6	4	6	25	3.1865	0	0
CaViaR	1	8	4	8	21	3.2381	0	0
CaViaR-EVT	10	9	8	9	36	3.256	1	1
AL-CaViaR	3	10	3	10	26	3.2334	0	0
Panel B: $\alpha = 0.975$								
Risk Model	McNeil Bootstrap	ESR Two-Sided	ESR One-Sided	ESR Auxiliary	Total	$S(VaR, ES, y, \alpha)$	MCS 75%	MCS 90%
GQRF	1	9	1	10	21	3.5681	0	0
GJR-GARCH Skew-t	10	7	8	7	32	3.5798	1	0
GARCH-EVT	5	9	5	7	26	3.5684	1	1
HAR Range-EVT	7	6	6	5	24	3.5196	0	0
CaViaR	0	9	3	8	20	3.5973	1	0
CaViaR-EVT	5	10	3	9	27	3.6143	1	0
AL-CaViaR	2	9	3	9	23	3.5855	0	0

re-balancing dates t and $t + h$. Following Happersberger et al. (2020), we do not allow for short positions. As such, the exposure to the risky asset E_t used to construct the portfolio is

$$E_t = \max [\min(C_t|\hat{\rho}_t(Y_t|X_t)|^{-1}, V_t, 0)] \quad (16)$$

where $\hat{\rho}_t(\cdot)$ is a VaR or ES risk forecast for the return Y_t using the predictor set X_t . We use a quarterly investment horizon to set the portfolio value floor, since this is generally the horizon over which an institutional investor makes their dynamic asset allocation decisions. Moreover, using a quarterly investment horizon as opposed to a yearly investment horizon reduces the so-called “lock-in” effect associated with portfolio insurance strategies whereby investors must wait until the next investment period if they fully exhaust their risk capital. A risk model that successfully captures large draw-downs and quickly adjusts to normalizations in market volatility will generate superior risk adjusted returns and lower downside risk in the DPPI strategy. We use the 99% VaR and 97.5% ES forecasts in our DPPI strategies as these are the levels of confidence that are of interest to regulators.

We run the DPPI strategy for each of the 10 stocks in our sample and use the Vanguard U.S. total bond market ETF as the low-risk asset. To evaluate investment performance we focus on reporting measures of downside risk and risk-adjusted return. In particular, we report the Sharpe ratio which is defined as the ratio of the excess⁹ return divided by the standard deviation, the Sortino ratio which is defined as the ratio of

⁹We always define excess returns as the strategy return above the return on cash.

the excess return divided by the standard deviation of only negative returns (downside deviation), the MAR ratio which divides the strategy’s compound annual growth rate by the strategy’s maximum drawdown, and the Omega ratio which is the ratio of the expectations of gains to the expectation of losses using a return threshold of zero (Bertrand and Prigent, 2011). We also report the maximum drawdown, maximum drawdown per unit of volatility, and the annualized return and standard deviation. All strategy performance statistics are computed net of transaction costs which are assumed to be 0.3% of the value traded (Petraki, 2020). We also include a cost equal to 0.1% of the prevailing mid-price to account for slippage at the time of the transaction. To mitigate path dependence, that is the dependence of the strategy returns on a given starting date, we average the investment performance measures over all possible starting days between 2010-12-22 and 2011-01-19 and all stocks. The exception is the maximum drawdown statistic, where we take the maximum instead of the average. The backtests end on 2019-11-29. The portfolio re-balancing period is set to match the forecast horizon.

Table 6 and Table 7 report performance statistics for the DPPI investment strategies using the one-day ahead and 10-day ahead VaR/ES forecasts respectively. At the one-day ahead frequency we find that the GQRF model has a higher Sortino ratio than the CaViaR and CaViaR EVT models when using 99% VaR forecasts, and a higher Sortino ratio than all benchmark models when using the 97.5% ES forecasts. At the 10-day ahead forecast horizon, our GQRF model has a higher Sortino ratio than all benchmark models except the GARCH-EVT model.

The maximum drawdowns associated with our GQRF model strategy are smaller than most benchmark model strategies at the one-day ahead horizon, particularly when using the 97.5% ES forecasts. At the 10-day ahead horizon only the GARCH-EVT model strategy has a marginally smaller maximum drawdown. When considering the MAR ratio, the GQRF model tends to outperform most benchmark models over all horizons and risk measures.

Concerning the Omega ratio at the one-day ahead forecast horizon, we find that the GQRF model outperforms the three CaViaR type models for the 99% VaR, and all benchmark models for the 97.5% ES. At the 10-day ahead forecast horizon the Omega ratio of our GQRF model DPPI strategy is only beaten by the GARCH-EVT model. Finally, we note that our GQRF model earns a highly competitive Sharpe ratio at both forecast horizons, particularly when compared against the benchmark semi-parametric CaViaR type models. In summary, while the aforementioned differences in performance statistics from the DPPI investment strategies are not large, we do find that our GQRF risk model can outperform benchmark risk models net of transaction costs in a realistic portfolio insurance strategy, particularly at the 10-day ahead forecast horizon. Among benchmark models, the DPPI strategies that use forecasts from the GARCH and GARCH-EVT models perform the best when considering all performance metrics and risk levels. Our GQRF risk model always earns higher Sharpe, Sortino, and Omega ratios at the 10-day forecast horizon relative to the one-day ahead horizon for both the 99% VaR and 97.5% ES risk measures. This is not necessarily the case for the benchmark risk models that we study and suggests that our GQRF model works well for forecasting financial risk over longer horizons, a result that is particularly important in practice.

Table 6: One-day Ahead DPPI Strategy Results

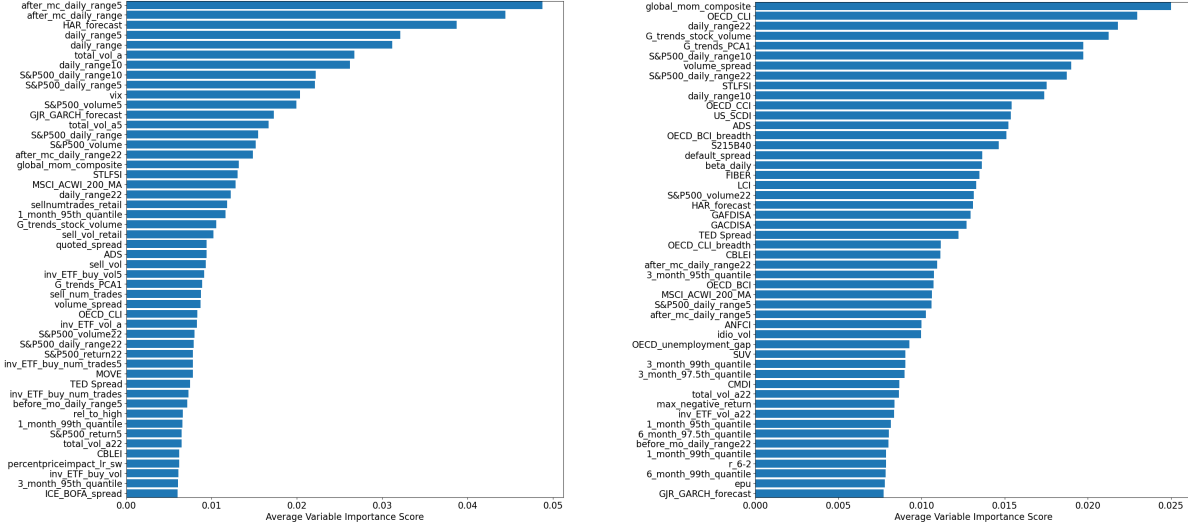
The table reports the average investment performance measures from the DPPI investment strategy backtest using one-day ahead 99% VaR and 97.5% ES forecasts. All measures are net of transaction costs.

Panel A: 99% VaR							
	GQRF	GARCH	GARCH EVT	HAR Range-EVT	CaViaR	CaViaR-EVT	AL-CaViaR
Annualised Return (%)	4.2478	4.6611	4.1587	4.0652	4.0645	3.5887	3.9897
Annualised Std	7.3151	7.2581	6.9036	6.8171	7.0700	6.6875	6.8316
Annualised Sharpe Ratio	0.5560	0.6062	0.5866	0.5778	0.5421	0.5122	0.5548
Annualised Sortino Ratio	0.8849	0.9762	0.9422	0.9207	0.8653	0.8115	0.8924
Max Drawdown (%)	-13.1519	-13.3391	-12.5791	-12.3663	-13.5431	-13.9426	-13.3175
MAR Ratio	0.3495	0.3558	0.3485	0.3536	0.3552	0.3092	0.3440
Max DD/Vol	1.8072	1.8403	1.7721	1.7769	1.9121	1.9225	1.8894
Omega Ratio	1.1320	1.1427	1.1369	1.1328	1.1272	1.1182	1.1289
Panel B: 97.5% ES							
	GQRF	GARCH	GARCH EVT	HAR Range-EVT	CaViaR	CaViaR-EVT	AL-CaViaR
Annualised Return (%)	4.4013	4.4596	4.2409	4.2982	4.2006	4.0672	3.7547
Annualised Std	6.3827	7.3050	6.6806	6.7384	6.5940	7.2316	6.6278
Annualised Sharpe Ratio	0.6595	0.5801	0.6311	0.6160	0.6130	0.5517	0.5434
Annualised Sortino Ratio	1.0819	0.9337	1.0203	0.9921	1.0067	0.8738	0.8838
Max Drawdown (%)	-10.6677	-13.4913	-11.8500	-11.9684	-12.3522	-13.9591	-12.1503
MAR Ratio	0.4508	0.3356	0.4227	0.3822	0.3895	0.3246	0.3360
Max DD/Vol	1.6628	1.8534	1.6939	1.7574	1.8176	1.9206	1.8326
Omega Ratio	1.1519	1.1366	1.1463	1.1419	1.1422	1.1293	1.1255

Table 7: 10-day Ahead DPPI Strategy Results

The table reports the average investment performance measures from the DPPI investment strategy backtest using 10-day ahead 99% VaR and 97.5% ES forecasts. All measures are net of transaction costs.

Panel A: 99% VaR							
	GQRF	GARCH	GARCH EVT	HAR Range-EVT	CaViaR	CaViaR-EVT	AL-CaViaR
Annualised Return (%)	4.1840	4.3106	4.1673	4.4134	4.1303	4.1504	4.2111
Annualised Std	4.0918	4.7805	3.9571	4.4203	4.5231	4.2921	4.4655
Annualised Sharpe Ratio	1.0508	0.9369	1.0918	1.0210	0.9665	0.9934	0.9922
Annualised Sortino Ratio	1.7504	1.5465	1.8272	1.6965	1.6085	1.6434	1.6494
Max Drawdown (%)	-11.4901	-14.1474	-9.9884	-12.8944	-15.3149	-13.3877	-14.0649
MAR Ratio	0.6698	0.6204	0.7039	0.6610	0.6388	0.6393	0.6553
Max DD/Vol	1.7977	1.8183	1.7720	1.7706	1.8794	1.8193	1.8507
Omega Ratio	1.2117	1.1972	1.2168	1.2109	1.1996	1.2033	1.2043
Panel B: 97.5% ES							
	GQRF	GARCH	GARCH EVT	HAR Range-EVT	CaViaR	CaViaR-EVT	AL-CaViaR
Annualised Return (%)	4.1825	4.3040	4.1441	4.4117	4.1648	4.2100	4.2081
Annualised Std	4.0769	4.8015	3.9421	4.3865	4.1302	4.4487	4.2227
Annualised Sharpe Ratio	1.0502	0.9338	1.0926	1.0297	1.0415	0.9782	1.0267
Annualised Sortino Ratio	1.7487	1.5414	1.8284	1.7129	1.7343	1.6170	1.7135
Max Drawdown (%)	-10.8880	-14.4630	-9.8058	-12.7299	-11.9004	-13.9374	-13.1253
MAR Ratio	0.6698	0.6159	0.7000	0.6657	0.6695	0.6354	0.6680
Max DD/Vol	1.7986	1.8276	1.7789	1.7748	1.8493	1.8394	1.8465
Omega Ratio	1.2123	1.1967	1.2165	1.2123	1.2104	1.2019	1.2090



(a) One-day ahead 99% VaR Model

(b) 10-day ahead 99% VaR Model

Figure 1: Variable importance scores from the 99% VaR GQRF models

6 Variable Importance

Our GQRF model facilitates an analysis of the relative importance of each predictor variable for VaR and ES forecasting. This analysis provides an insight into which variables are most strongly related to future financial risk in a non-parametric framework. Furthermore, the variable importance scores from our GQRF model could guide regulators in selecting variables that can be used to monitor financial risk ex-ante and to design realistic stress testing scenarios.

A variable importance score from our GQRF model is computed as the weighted sum of how many times each variable was chosen for splitting in the forest at each forecast time. As such, higher scores indicate more important variables. Figure 1 and Figure 2 plot the variable importance scores averaged over stocks and out-of-sample forecast dates for the top 50 variables at both one-day ahead and 10-day ahead forecast horizons for the $\alpha = 0.99$ VaR models and the $\alpha = 0.975$ ES models respectively. We focus on these two specific levels of risk because they are the levels that are most important to regulators. At the $h = 1$ forecast horizon, the top 10 most important variables in our GQRF model are predominately measures of volatility, both stock-specific volatility (e.g., **after_mc_daily_range** and **daily_range**) and volatility for the market index (S&P500). This finding may explain why prior literature has predominantly focused on incorporating volatility measures into financial risk forecast models (see, e.g., Gerlach and Wang, 2020). However, our model extends this literature by allowing for interactions and non-linearity when volatility is used to model financial risk.

We find that measures of volatility during the after-market close trading period are particularly important for forecasting the one-day ahead VaR and ES. Moreover, total trade volume in the after-market close

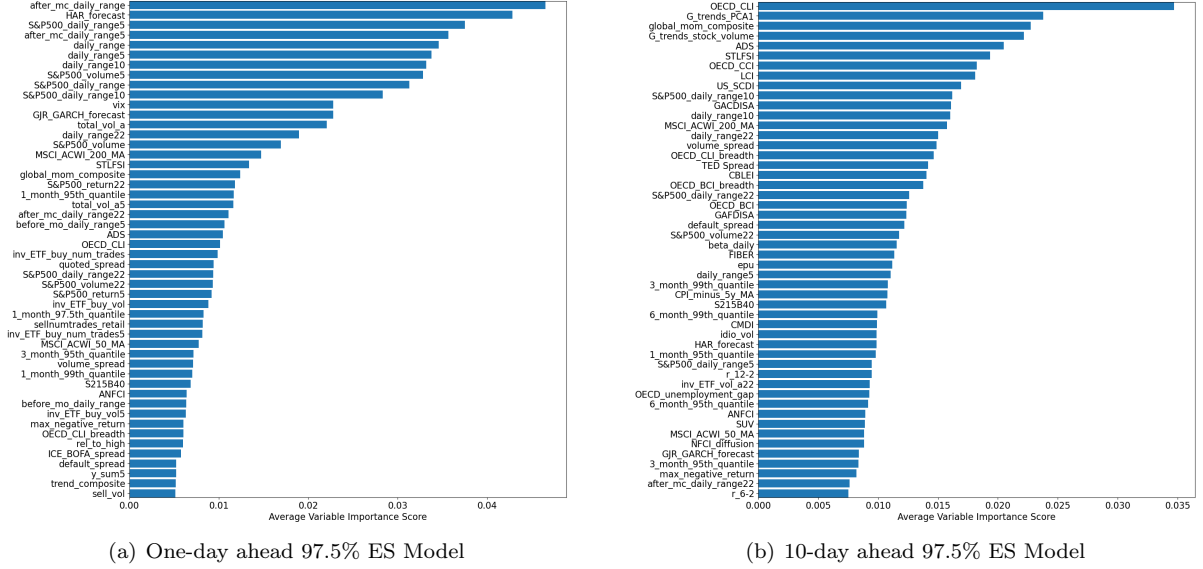
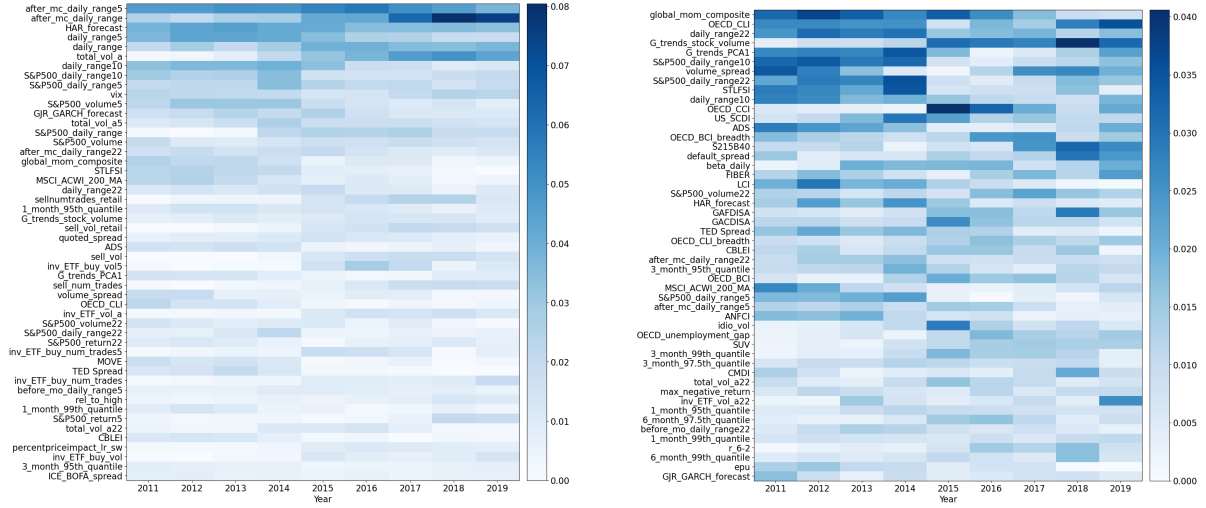


Figure 2: Variable importance scores from the 97.5% ES GQRF models

trading session is also selected as a highly important variable by our GQRF risk model. To the best of our knowledge, prior literature has not widely considered the role of after-market trading data for financial risk forecasting applications. The second most important group of variables are measures of global equity market momentum and price trends (e.g., the **global_momentum_composite** and the **MSCIACWI.200_MA** variables), financial stress indices (e.g., the **STLSFI** and **ADS** indices), and measures of trading activity and the market microstructure (e.g., **sellvol_retail**, **quoted_spread**). We also find that the features we engineer from Google Trends data, and features related to trading activity in the inverse S&P500 ETF are important. To the best of our knowledge, we are the first to demonstrate the utility of these types of variables for financial risk forecasting.

The dynamics of the variable importance scores differ substantially at the 10-day forecasting horizon. The difference between the score for the most important variable and the score least important variable is considerably smaller at the 10-day ahead horizon than for the one-day ahead horizon. Moreover, the variable importance scores decay at a slower rate. This finding suggests that our GQRF risk model uses a wider variety of information to make accurate 10-day ahead VaR and ES forecasts. The 10 most important variables at this horizon are now a mix of volatility variables, financial stress indices, and equity market momentum and breadth variables. In general, we find that volatility variables and retail trading activity variables are considerably less important for longer horizon VaR and ES forecasting in our GQRF model. This may reflect the idea that retail trading activity is uninformative for understanding the tail dynamics of stock returns over the longer term. The variables engineered using the OECD composite leading indicator data, the OECD business confidence index, and the OECD consumer confidence index, are particularly important for 10-day ahead risk forecasting. Moreover, the two variables engineered from Google Trends



(a) One-day ahead 99% VaR Model

(b) 10-day ahead 99% VaR Model

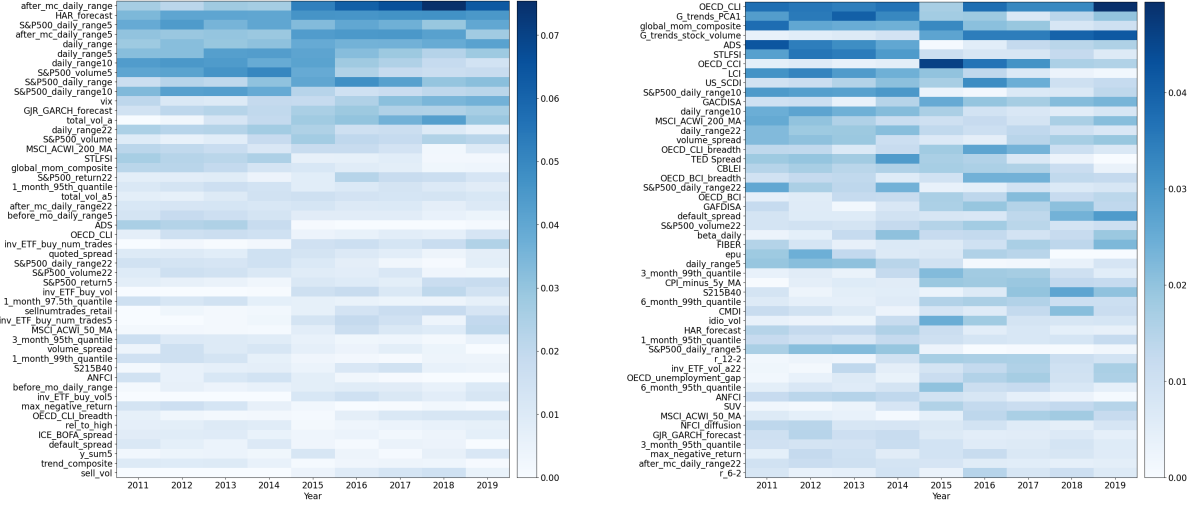
Figure 3: Yearly variable importance scores from the 99% VaR GQRF models from 2011 - 2019. Darker cells indicate larger variable importance scores.

data are ranked higher in the 10-day ahead risk forecasting model compared to the one-day ahead forecasting model. These results further highlight the importance of alternative data sources for modern financial risk management.

To study the dynamic importance of each variable in our GQRF risk model, Figure 3 and Figure 4 plot the mean importance score in each year of our sample averaged over all stocks at both the one-day ahead and 10-day ahead horizons for the $\alpha = 0.99$ VaR model and the $\alpha = 0.975$ ES model respectively. At the one-day ahead horizon we find that most volatility variables are important over our entire backtest sample. This result further explains why prior literature has predominately focused on including only realized volatility measures in risk forecasting models. For both the one-day ahead and 10-day ahead horizon, variables that measure retail trading activity and trading activity in inverse ETFs tend to become more important toward the end of our sample. This dynamic potentially reflects the growing importance of retail trading activity on stock price movements (see, e.g., Boehmer et al., 2021; Han and Kumar, 2013). The emerging importance of retail and inverse ETF trading activity data that we uncover could be of particular interest to regulators and practitioners who wish to identify new variables that may help to anticipate short-term downside risk.

At both forecasting horizons financial stress indices and the Google Trends variable engineered from a list of search terms about financial risk tend to be more important towards at the beginning of our backtest. This is likely because the training sample used to produce the out-of-sample forecasts contains the Global Financial Crisis period. In contrast, the variable engineered from Google Trends search volume about the individual stock tends to become increasingly important during the latter years of our backtest sample.

Finally, we note that the log change in the OECD CLI indicator is always an important variable at the 10-day ahead horizon over all years in our backtest for both the VaR and ES. Practitioners may wish to



(a) One-day ahead 97.5% ES Model

(b) 10-day ahead 97.5% ES Model

Figure 4: Yearly variable importance scores from the 97.5% ES GQRF models from 2011 - 2019. Darker cells indicate larger variable importance scores.

consider this variable when monitoring longer-term financial risk. Figure 1 - Figure 4 show that the top 50 most important variables differ depending upon the forecast horizon and vary dynamically through time. Our results therefore underscore the importance of incorporating adaptive variable selection methods into financial risk forecasting models that use a large number of predictor variables, and highlight the dynamic importance of idiosyncratic and market wide information for financial risk forecasting.

7 Conclusion

The AMH postulates that the degree of market efficiency varies through time as investors adapt to prevailing market conditions and changing informational environments (Lo, 2004, 2012). When market prices do not fully reflect available information additional predictor variables may be required to adequately forecast financial risk. This study designs a machine learning based financial risk forecasting model that is built using Generalized Quantile Random Forests. Our non-parametric model can exploit a large number of predictor variables, naturally allows for variable interactions, and can model non-linear relationships to forecast the h -period VaR and ES. As such, our risk model aligns with the implications of the AMH. To limit the risk of overfitting the noisy financial return data and to ensure that our model is sufficiently generalizable we have employed two approaches, namely a model-free quantile adaptive screening procedure and a robust cross-validation technique.

Using over 100 economic and financial variables that constitute a diverse source of information we demonstrate that our GQRF risk model produces reliable and competitive VaR and ES forecasts at both the one-day

ahead and 10-day ahead horizon across multiple levels of confidence. When used in a dynamic proportion portfolio insurance investment strategy the VaR and ES forecasts from our GQRF model generate returns with attractive levels of downside risk relative to benchmark models, particularly at the 10-day ahead forecast horizon. For short-term risk forecasts, we find that measures of volatility are particularly important, including measures of volatility during the after-market close trading session. At the 10-day ahead horizon leading economic indicators, financial market stress indices, and equity market momentum and breadth variables are important for forecasting the VaR and ES. In addition, we demonstrate that variables engineered from Google Trends search data, retail investor trading activity, trading activity in inverse ETFs, and OECD leading indicators play an important role in our non-parametric GQRF risk model. These results may be of particular importance to regulators and practitioners who seek to identify variables that can be used to anticipate future financial risk. Future research may wish to further explore how to effectively shrink the size of the predictor set while retaining the most important predictive information within the context of our GQRF risk forecasting model¹⁰.

Our study contributes to the literature by enabling the use of a large number of variables to forecast the VaR and ES. This addresses substantial limitations of existing financial risk forecasting models. Overall, we have demonstrated the utility of applying a non-parametric machine learning technique to a rich dataset of relevant economic and financial variables to improve financial risk management practices.

¹⁰We thank an anonymous referee for this suggestion.

A Benchmark Risk Models

A.1 GARCH-EVT

The two-step GARCH-EVT model of McNeil and Frey (2000) first involves fitting a GARCH model to the raw returns, then applying “Peaks-Over-Threshold” (POT) extreme value theory to approximate the tail of standardized residuals from the GARCH model. We use a GJR-GARCH model, since Trapin (2018) demonstrate that volatility models endowed with the so called “leverage” effect better remove the dependence among the most extreme negative returns and produce innovations that are closer to i.i.d. The GJR-GARCH specification is as follows,

$$Y_t = \mu + \phi_1 Y_{t-1} + \epsilon_t \quad (17)$$

$$v_t^2 = \omega + \alpha_1 \epsilon_{t-1}^2 + \gamma_1 \epsilon_{t-1}^2 \mathbb{1}(\epsilon_{t-1} < 0) + \psi_1 v_{t-1}^2 \quad (18)$$

$$\epsilon_t = v_t e_t \quad (19)$$

Model parameter are estimated by maximum likelihood.

An initial threshold value, denoted by u , is set at the 90th quantile of the distribution of standardized residuals. The set of excess residuals over this threshold is $Z = \{\epsilon_t - u : \epsilon_t \geq u\}$. The Pickands-Balkema-de Haan theorem states that the distribution of the excess loss converges to the generalized Pareto family of distributions for a suitably large threshold value, (Pickands III et al., 1975; Balkema and De Haan, 1974). The generalized Pareto distribution function with shape ξ and scale σ is

$$G_{\xi, \sigma}(z) = \begin{cases} 1 - \left(1 + \frac{\xi z}{\sigma}\right)^{-1/\xi}, & \text{for } \xi \neq 0, z > 0 \\ 1 - \exp\left(-\frac{z}{\sigma}\right), & \text{for } \xi = 0, z > 0 \end{cases} \quad (20)$$

When $\xi > 0$ the generalized Pareto distribution is heavy-tailed. Parameters of this distribution are estimated using maximum likelihood. Assuming that the tail of F begins at u , then for some $\eta > u$ the following equality holds,

$$F(\eta) = 1 - (1 - F(u))(1 - F_u(\eta - u)), \quad \eta > u \quad (21)$$

where F_u represents the distribution of the excess loss above u . The POT method constructs an estimate for $F(\eta)$. The term $(1 - F(u))$ is approximated non-parametrically by the proportion of observations in the tail region above u , that is N_u/T . The distribution of the excess loss, $F_u(\eta - u)$, is approximated by the generalized Pareto distribution. Hence, the POT tail estimator is

$$\hat{F}_\eta(\eta) = 1 - \frac{N_u}{T} \left(1 + \frac{\xi \eta - u}{\sigma}\right)^{-1/\xi}. \quad (22)$$

Assuming that $\alpha > F(u)$, we may invert $\hat{F}_\eta$ to estimate the tail quantiles as,

$$\hat{F}_\eta^{-1}(\alpha) = u + \frac{\hat{\sigma}}{\hat{\xi}} \left[\left(\frac{T}{N_u}(1 - \alpha) \right)^{-\hat{\xi}} - 1 \right] \quad (23)$$

If $\xi < 1$, then an expression for the conditional mean beyond the α -quantile, that is the ES, can be derived. For some $u' > u$, the excess loss $\{Y - u' | Y > u'\}$ also follows a generalized Pareto distribution with an identical shape parameter ξ but new scale parameter given by $\sigma + \xi(u' - u)$ (Coles, 2001). Using the expression for the mean of the generalized Pareto distribution, we can write

$$\mathbb{E}[\epsilon | \epsilon > \hat{F}_\eta^{-1}(\alpha)] = \frac{\hat{F}_\eta^{-1}(\alpha) + \sigma - \xi u}{1 - \xi}, \quad \xi < 1, \alpha > F(u). \quad (24)$$

Using the one-step ahead mean and one-step ahead variance forecasts from the GJR-GARCH model, the one-step ahead VaR and ES forecasts from the GARCH-EVT model are computed as

$$\widehat{VaR}_{t+1}^{(\alpha)} = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{F}_\eta^{-1}(\alpha) \quad (25)$$

$$\widehat{ES}_{t+1}^{(\alpha)} = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \mathbb{E}[\epsilon | \epsilon > \hat{F}_\eta^{-1}(\alpha)] \quad (26)$$

To compute the multi-step ahead VaR and ES forecasts we follow the simulation algorithm proposed by McNeil and Frey (2000) using 1000 future return paths.

A.2 CaViaR, CaViaR-EVT and AL-CaViaR

The conditional autoregressive Value-at-Risk model of Engle and Manganelli (2004a), uses quantile regression to estimate the VaR. We focus on the asymmetric slope (AS) CaViaR model (CaViaR henceforth) since this is the most widely used within the literature. The CaViaR model is

$$\widehat{VaR}_{t+1}^{(\alpha)} = \beta_0 + \beta_1 \widehat{VaR}_t^{(\alpha)} + \beta_2 (\tilde{y}_t)^+ + \beta_3 (\tilde{y}_t)^- \quad (27)$$

where $\tilde{y}$ is the de-meaned log return series, $(x)^+ = \max(x, 0)$ and $(x)^- = -\min(x, 0)$. The parameters of the CaViaR model are obtained as the solution to the following optimization problem

$$\beta = \arg \min_{\beta} \sum_{t=1}^T (\alpha - \mathbb{1}(\tilde{y}_{t+1} < \widehat{VaR}_{t+1}^{(\alpha)})) (\tilde{y}_{t+1} - \widehat{VaR}_{t+1}^{(\alpha)}). \quad (28)$$

The CaViaR model only produced an estimate of the VaR. We use the mean exceedance method to calculate an estimate of the ES (see, e.g., Taylor, 2019). Specifically, the ES is calculated as the sum of the VaR and the average of all in-sample returns that exceed the estimated VaR.

To estimate the CaViaR-EVT model we fit the CaViaR model for $\alpha = 0.925$ to obtain a time series of conditional quantile estimates and define $z = y_{t+1} / \widehat{VaR}_{t+1}^{(0.925)} - 1$. The POT extreme value theory method

detailed in Appendix A.1 is then applied to this sequence to estimate the quantile and conditional tail expectation used to forecast the one-day ahead VaR and ES.

The AL-CaViaR model introduced by Taylor (2019) extends the CaviaR model to jointly estimate the VaR and the ES. The specification for the VaR is identical to (27) and the ES is specified as

$$\widehat{ES}_{t+1}^{(\alpha)} = (1 + \exp(\gamma)) \widehat{VaR}_{t+1}^{(\alpha)} \quad (29)$$

This is the simplest specification for the ES considered by (Taylor, 2019) and was shown to perform particularly well in backtests. The parameters of the AL-CaViaR model are found by maximum likelihood of the Asymmetric Laplace density. The Asymmetric Laplace density defined by Taylor (2019) is as follows

$$f(y_{t+1}) = \frac{(\alpha - 1)}{ES_{t+1}^{(\alpha)}} \frac{(\alpha - \mathbb{1}(\tilde{y}_{t+1} < VaR_{t+1}^{(\alpha)}))(\tilde{y}_{t+1} - VaR_{t+1}^{(\alpha)})}{\alpha ES_{t+1}^{(\alpha)}} \quad (30)$$

To compute the multi-step ahead VaR and ES forecasts we use the modified square-root of time scaling rule proposed by Wang et al. (2011). This approach multiplies the 1-day ahead risk forecasts from the CaViaR and CaViaR-EVT models by $\sqrt{h \times VR(h)}$, where $VR(h)$ is the estimated variance ratio statistic of Lo and MacKinlay (1988). Wang et al. (2011) argue that this modified square root of time scaling rule corrects for the downward bias associated with the original square root of time scaling rule when returns exhibit temporal dependence.

To set the starting parameters for the numerical minimization routine we follow the approach taken by Taylor (2008). For the forecast date associated with the first moving window of training data in the backtest we use 10^5 different parameter vectors of uniformly distributed random numbers in the half-open interval $[0,1)$ as starting values to obtain multiple different solutions. The 10 vectors of starting values that produced the smallest value of the quantile regression objective function are retained. These solutions obtained from the 10 best vectors of starting values are then re-used as the initial values in the numerical minimization routine and we select the single parameter vector which minimizes the quantile regression objective function as the final set of starting parameters. For all subsequent forecast dates in the backtest we use the parameter values obtained as the solution for the previous forecasting period's moving window as starting values.

A.3 HAR Range-EVT

The HAR-Range EVT models the conditional volatility of returns using the Heterogeneous auto-regressive (HAR) model of Corsi (2009) and applies POT extreme value theory to the set of standardized residuals (see, e.g., Trapin, 2018). The HAR-Range model is built using a daily range volatility estimator. Following Molnár (2012) and Lyócsa et al. (2021), we use the daily range estimator of Garman and Klass (1980) which also incorporates the overnight jump component, that is

$$GK_t = \frac{1}{2} \log \left(\frac{H_t}{L_t} \right)^2 - (2 \log(2) - 1) \log \left(\frac{C_t}{O_t} \right)^2 + \log \left(\frac{O_t}{C_{t-1}} \right)^2 \quad (31)$$

where H_t is the daily high price and L_t is the daily low price, C_t is the daily close price, O_t is the daily open price. The use of the daily range volatility estimators is consistent with the volatility estimators used in our GQRF risk model. This estimator is simple to compute and uses publicly available price data, which is particularly important from a practical perspective.

The HAR model is expressed as a linear function of the past daily, weekly ($GK_t^{(w)} = \frac{1}{5} \sum_{i=1}^5 GK_{t-i+1}$), and monthly ($GK_t^{(m)} = \frac{1}{22} \sum_{i=1}^{22} GK_{t-i+1}$) daily range volatility components. The HAR Range model is

$$GK_{t+1} = \beta_0 + \beta_1 GK_t + \beta_2 GK_t^{(w)} + \beta_3 GK_t^{(m)} + \epsilon \quad (32)$$

Following Clements and Preve (2021), we estimate the parameters of the HAR Range model using weighted least squares, where the weights are defined as $1/GK_t$. Then we apply the standard POT framework outlined in Appendix A.1 to the sequence of negative returns that have been standardized using the sample mean and HAR Range model conditional variance estimates to forecast the one-day ahead VaR and ES.

To compute multi-step ahead forecasts we forecast the multi-step ahead variance from the HAR Range model by iterating equation (32) (see, e.g., Chiriac and Voev, 2011), and apply the simulation algorithm of McNeil and Frey (2000). In this approach we simulate 1000 future return paths and assume a Student-t distribution for the filtered returns.

B Predictor Variables

Table 8: Variable Descriptions

The Table describes briefly each variable included in the GQRF risk model and provides the data source.

Acronym	Description	Freq	Data Source
(i) Past Returns/Prices			
r_12-2	Cumulative log return from 12 months before to 2 months before the forecast date.	D	Compustat
r_6-2	Cumulative log return from 6 months before to 2 months before the forecast date.	D	Compustat
S&P500_return5	Cumulative S&P500 return over the past 5 days.	D	Compustat
S&P500_return22	Cumulative S&P500 return over the past 22 days.	D	Compustat
6_month_95th_quantile	The 95th empirical quantile of negative log returns over the past 6 months.	D	Compustat
3_month_95th_quantile	The 95th empirical quantile of negative log returns over the past 3 months.	D	Compustat
1_month_95th_quantile	The 95th empirical quantile of negative log returns over the past month.	D	Compustat
6_month_97.5th_quantile	The 97.5th empirical quantile of negative log returns over the past 6 months.	D	Compustat
3_month_97.5th_quantile	The 97.5th empirical quantile of negative log returns over the past 3 months.	D	Compustat
1_month_97.5th_quantile	The 97.5th empirical quantile of negative log returns over the past month.	D	Compustat
6_month_99th_quantile	The 99th empirical quantile of negative log returns over the past 6 months.	D	Compustat
3_month_99th_quantile	The 99th empirical quantile of negative log returns over the past 3 months.	D	Compustat
1_month_99th_quantile	The 99th empirical quantile of negative log returns over the past month.	D	Compustat
rel.to_high	Price relative to the rolling 52-week high.	D	Compustat
max_negative_return	The largest negative return over the previous 22 trading days.	D	Compustat
beta_daily	The sum of regression coefficients from a regression of daily excess returns on the contemporaneous and lagged market excess return.	D	Compustat
lag_y	The lagged negative log return.	D	Compustat
lag_y5	The 5-day lagged negative log return.	D	Compustat

Acronym	Description	Freq	Data Source
lag_y10	The 10-day lagged negative log return.	D	Compustat
lag_y22	The 22-day lagged negative log return.	D	Compustat
y_sum5	Cumulative sum of negative log returns over the past 5 trading days.	D	Compustat
y_sum10	Cumulative sum of negative log returns over the past 10 trading days.	D	Compustat
y_sum22	Cumulative sum of negative log returns over the past 22 trading days.	D	Compustat
global_mom_composite	A composite index of momentum variables for all MSCI ACWI markets.	D	NDR
MSCI_ACWI_200_MA	Percentage of MSCI ACWI stocks above their 200 moving average prices.	D	NDR
trend_composite	A composite index of price trend indicators for the U.S market.	D	NDR
MA_composite	A composite index of moving average variables for the U.S market.	D	NDR
S215B40	% of NDR All Cap Stocks Above 40-Week Moving Averages.	D	NDR
S215B10	% of NDR All Cap Stocks Above 10-Week Moving Averages.	D	NDR
I5210	% of MSCI ACWI markets above their 50-day MA.	D	NDR
MSCI_ACWI_50_MA	Percentage of MSCI ACWI stocks above their 50 moving average price.	D	NDR
rsi	The Relative Strength Index.	D	Compustat
(ii) Volatility			
vix	The VIX volatility index.	D	FRED
MOVE	The Merrill Lynch Option Volatility Estimate Index.	D	Bloomberg
daily_range	The daily range volatility estimator of Garman and Klass (1980).	D	Compustat
daily_range5	The daily range volatility estimator of Garman and Klass (1980) averaged over the past 5 trading days.	D	Compustat
daily_range10	The daily range volatility estimator of Garman and Klass (1980) averaged over the past 10 trading days.	D	Compustat
daily_range22	The daily range volatility estimator of Garman and Klass (1980) averaged over the past 22 trading days.	D	Compustat
after_mc_daily_range	The high/low range volatility estimator of Parkinson (1980) for the after market close trading period.	D	NYSE TAQ

Acronym	Description	Freq	Data Source
after_mc_daily_range5	The high/low range volatility estimator of Parkinson (1980) for the after-market close trading period averaged over the past 5 trading days.	D	NYSE TAQ
after_mc_daily_range22	The high/low range volatility estimator of Parkinson (1980) for the after-market close trading period averaged over the past 22 trading days.	D	NYSE TAQ
before_mo_daily_range	The high/low range volatility estimator of Parkinson (1980) for the pre-opening session.	D	NYSE TAQ
before_mo_daily_range5	The high/low range volatility estimator of Parkinson (1980) for the pre-opening session averaged over the past 5 trading days.	D	NYSE TAQ
before_mo_daily_range22	The high/low range volatility estimator of Parkinson (1980) for the pre-opening session averaged over the past 22 trading days.	D	NYSE TAQ
GJR_GARCH_forecast	h -step ahead volatility forecast from a GJR GARCH model with skew-t errors. The GARCH model parameters are estimated using a rolling 1500-day sample.	D	Compustat
HAR_forecast	h -step ahead volatility forecast from a HAR model (see, e.g., Corsi and Renò, 2012). The HAR model parameters are estimated using robust regression with a Tukey bi-weight kernel as in Clements and Preve (2021), using a rolling 1500-day sample.	D	Compustat
S&P500_range	The daily range volatility estimator of Garman and Klass (1980) for the S&P500 index.	D	Compustat
S&P500_range5	The daily range volatility estimator of Garman and Klass (1980) for the S&P500 index averaged over the past 5 trading days.	D	Compustat
S&P500_range22	The daily range volatility estimator of Garman and Klass (1980) for the S&P500 index averaged over the past 22 trading days.	D	Compustat
idio_vol	Idiosyncratic volatility.	D	Compustat
vrp_30	Variance risk premia for 30-day S&P500 options.	D	OptionMetrics
CBOE_SKEW	Skewness of S&P 500 returns at the end of a 30-day horizon.	D	CBOE
CBOE_SKEW5	5-day moving average of the CBOE_SKEW variable.	D	CBOE
CBOE_SKEW22	22-day moving average of the CBOE_SKEW variable.	D	CBOE
krp_30	Kurtosis risk premia for 30-day S&P500 options.	D	OptionMetrics

Acronym	Description	Freq	Data Source
(iii) Liquidity			
total_vol_a	Trade volume during the after-market close session.	D	NYSE TAQ
total_vol_a5	Trade volume during the after-market close session averaged over the past 5 trading days.	D	NYSE TAQ
total_vol_a22	Trade volume during the after-market close session averaged over the past 22 trading days.	D	NYSE TAQ
Inv ETF_vol_a	Trade volume during the after-market close session for the ProShares Short S&P500 ETF.	D	NYSE TAQ
Inv ETF_vol_a5	The cumulative sum of trade volume during the after-market close session over the past 5 trading days for the ProShares Short S&P500 ETF.	D	NYSE TAQ
Inv ETF_vol_a22	The cumulative sum of trade volume during the after-market close session over the past 22 trading days for the ProShares Short S&P500 ETF.	D	NYSE TAQ
S&P500_volume	The volume of the S&P500 index.	D	Compustat
S&P500_volume5	The volume of the S&P500 index averaged over the past 5 trading days.	D	Compustat
S&P500_volume22	The volume of the S&P500 index averaged over the past 22 trading days.	D	Compustat
sellnumtrades_retail	Total number of retail sells.	D	NYSE TAQ
quoted_spread	The time-weighted quoted bid-ask spread.	D	NYSE TAQ
bid_depth	Time weighted depth at the best bid price	D	NYSE TAQ
sell_vol	Daily sell volume.	D	NYSE TAQ
sell_vol5	The cumulative sum of sell volume over the past 5 trading days.	D	NYSE TAQ
sell_vol22	The cumulative sum of sell volume over the past 22 trading days.	D	NYSE TAQ
sell_vol_retail	Daily retail sell volume.	D	NYSE TAQ
sell_num_trades	Total number of sells.	D	NYSE TAQ
sell_num_trades5	Cumulative sum of sells over the past 5 trading days.	D	NYSE TAQ
sell_num_trades22	Cumulative sum of sells over the past 22 trading days.	D	NYSE TAQ
percentpriceimpact_lr_sw	The share weighted percentage price impact.	D	NYSE TAQ
tsignsqtrdvol	The price impact coefficient (Kyles lambda).	D	NYSE TAQ
inv ETF_buy_vol	Buy volume on the ProShares Short S&P500 ETF.	D	NYSE TAQ
inv ETF_buy_vol5	The cumulative sum of buy volume over the past 5 trading days on the ProShares Short S&P500 ETF.	D	NYSE TAQ

Acronym	Description	Freq	Data Source
inv ETF_buy_vol22	The cumulative sum of buy volume over the past 22 trading days on the ProShares Short S&P500 ETF.	D	NYSE TAQ
inv ETF_buy_num_trades	The number of buys for the ProShares Short S&P500 ETF.	D	NYSE TAQ
inv ETF_buy_num_trades5	The cumulative sum of buys over the past 5 trading days on the ProShares Short S&P500 ETF.	D	NYSE TAQ
inv ETF_buy_num_trades22	The cumulative sum of buys over the past 22 trading days on the ProShares Short S&P500 ETF.	D	NYSE TAQ
volume_spread	BMES All-Cap daily advancing volume minus BMES All-Cap declining volume.	D	NDR
SUV	Standard unexplained volume	D	Compustat
(iv) Economic and Financial Conditions			
ANFCI	The adjusted Chicago Fed National Financial Conditions Index.	W	FRED
NFCI_diffusion	The Chicago Fed National Financial Conditions diffusion index.	W	FRED
STLFSI	The St. Louis Fed Financial Stress Index.	W	FRED
BBKMCOIX	The Brave-Butters-Kelly Coincident index	M	FRED
CBLEI	The annualized 6-month growth rate in the Conference Board's LEI.	M	Bloomberg
ADS	The Aruoba-Diebold-Scotti Business Conditions Index.	D	Philadelphia Fed
NFIB_optimism	The NFIB's Small Business Optimism Index.	M	NFIB
GSUSFCI	The YOY change in the Goldman Sachs Global Financial Conditions Index	D	Bloomberg
NFIB_breadth	The proportion of NFIB indicators with a positive change.	M	NFIB
S&P500_LI_index	The leading indicator index constructed by Ned Davis Research.	M	NDR
GACDISA	Current General Business Conditions index.	M	NY Fed
ESMSC_breadth	The proportion of Empire State Manufacturing Survey questions about current business conditions associated with a positive change.	M	NY Fed
GAFDISA	Future General Business Conditions index.	M	NY Fed
ESMSF_breadth	The proportion of Empire State Manufacturing Survey questions about future business conditions associated with a positive change.	M	NY Fed
OECD_CLI	The log change in the OECD Composite Leading Indicator for the U.S. in the previous month.	M	OECD

Acronym	Description	Freq	Data Source
OECD_BCI	The log change in the OECD Business Confidence Index for the U.S. in the previous month.	M	OECD
OECD_CCI	The log change in the OECD Consumer Confidence Index for the U.S. in the previous month.	M	OECD
OECD_BCI_breadth	The proportion of OECD countries with a positive log change in the leading Business Confidence Index over the previous month.	M	OECD
OECD_CLL_breadth	The proportion of OECD countries with a positive log change in the Composite Leading Indicator over the previous month.	M	OECD
OECD_CCI_breadth	The proportion of OECD countries with a positive log change in the Consumer Confidence Index over the previous month.	M	OECD
TED Spread	The spread between the 3-month LIBOR and the 3-month treasury bill rate.	D	FRED
default_spread	The spread between the Moody's Seasoned Baa Corporate Bond Yield and the Moody's Seasoned Aaa Corporate Bond Yield	D	FRED
ICE_BOFA_spread	The ICE BofA US High Yield Index Option-Adjusted Spread.	D	FRED
default_spread	The spread between Moody's AAA and BAA Corporate Bond Yield.	D	FRED
3y_tnote_26_wk_change	The 26-week change in the U.S. 3-year treasury note yield.	D	FRED
CPI_minus_5y_MA	U.S. CPI minus its 5-year moving average.	M	FRED
output_gap	Difference between the U.S. economy's potential GDP and the actual GDP.	M	NDR
CMDI	Corporate Bond Market Distress Index.	W	NY Fed
(v) Other			
G_trends_stock_volume	The log change in Google Trends search volume for the stock ticker over the previous month.	D	Google Trends
G_trends_PCA1	The first principle component of the log change in google trends volume for 66 search terms related to financial markets and financial risk	D	Google Trends
trading_sent_composite	Daily Trading Sentiment Composite index.	D	NDR
epu	The 5-day moving average of the Economic Policy Uncertainty Index.	M	FRED
BCO	Log change in the price of Brent Crude Oil.	D	FRED

Acronym	Description	Freq	Data Source
put_call_ratio	The ratio of total put option volume to total call option volume.	D	OptionMetrics

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